FINAL EXPLANATION: CORE MATHEMATICS GRADE 10

Student Assessment Document

FINAL EXPLANATION: CORE MATHEMATICS — GRADE 10 | SUB-STRAND 2.7: SURFACE AREA AND VOLUME OF SOLIDS	
Student Assessment Document	
Student Name	____
Class	____
Date	____

INSTRUCTIONS FOR STUDENTS
This is your Final Explanation document. Use it as a model to craft your own written answer to the driving question. Read each section carefully, study the exemplar responses, and then write your own explanation using evidence you gathered throughout the 8 lessons. Your answer should show that you understand the mathematics AND can apply it to real situations — just like an artisan in Eldoret designing metal containers for a posho mill. Write in full sentences. Show all working where calculations are required. Use correct mathematical vocabulary throughout.

Section 1: Unpacking the Phenomenon — Why the Mill Owner Was Puzzled	
Explain why the posho mill owner in Eldoret was confused. In your answer, clearly distinguish between surface area and volume, and explain why having the same surface area does NOT guarantee the same volume. Use the cylindrical drum and rectangular metal box as your examples.	The mill owner noticed something surprising: two containers made from the same total area of iron sheet held very different amounts of maize grain. To understand why, we need to separate two ideas that are easy to mix up — surface area and volume. Surface area is the total area of all the flat and curved faces that cover the outside of a solid. It tells you how much material — in this case, iron sheet — was used to build the container. Volume, on the other hand, is the amount of three-dimensional space inside the container. It tells you how much the container can hold — like how many kilograms of maize grain fit inside. The key insight is this: surface area and volume are related to the shape of a solid, but they do not change in the same way. When you change the shape of a container while keeping the surface area fixed, the volume can increase or decrease depending on the new shape. This means that two containers built from exactly the same area of iron sheet can have very different capacities. In our case, the cylindrical drum and the rectangular box both have the same surface area (same iron sheet used), but the cylinder encloses more interior space — so it holds more maize. The mill owner assumed equal material meant equal capacity, but shape is the hidden variable that changes everything.

Section 2: Calculating Surface Area — Showing Your Working	
A metalsmith in Eldoret uses a	To compare how much iron sheet each container uses, we calculate the total surface area of each solid.

sheet of iron to make two containers: a closed cylinder with radius 35 cm and height 80 cm, and a closed rectangular box with length 60 cm, width 50 cm, and height 70 cm. Calculate the total surface area of each container. Show all working and state which container used more iron sheet. Use $\pi = 22/7$.	▸ CYLINDER (radius r = 35 cm, height h = 80 cm) Total Surface Area of a cylinder = $2\pi r^2 + 2\pi rh$ $= 2 \times (22/7) \times 35^2 + 2 \times (22/7) \times 35 \times 80$ $= 2 \times (22/7) \times 1\,225 + 2 \times (22/7) \times 2\,800$ $= 2 \times 3\,850 + 2 \times 8\,800$ $= 7\,700 + 17\,600$ $= 25\,300$ cm² ▸ RECTANGULAR BOX (l = 60 cm, w = 50 cm, h = 70 cm) Total Surface Area = $2(lw + lh + wh)$ $= 2(60\times50 + 60\times70 + 50\times70)$ $= 2(3\,000 + 4\,200 + 3\,500)$ $= 2 \times 10\,700$ $= 21\,400$ cm² The cylinder uses more iron sheet (25 300 cm² > 21 400 cm²). This tells us that if a metalsmith wanted to use the same iron sheet for both, they would need to adjust the dimensions. Surface area calculations like these help artisans plan how much material to buy before cutting any metal — saving both time and money at the workshop.

Section 3: Calculating Volume — How Much Grain Can Each Container Hold?	
Using the same two containers from Section 2, calculate the volume of each solid. Show all working. Then explain what your results tell the posho mill owner about which container to choose for storing the most maize grain. Use $\pi = 22/7$.	Now we calculate how much maize grain each container can hold by finding its volume. ▸ CYLINDER (r = 35 cm, h = 80 cm) Volume of a cylinder = $\pi r^2 h$ $= (22/7) \times 35^2 \times 80$ $= (22/7) \times 1\,225 \times 80$ $= (22/7) \times 98\,000$ $= 22 \times 14\,000$ $= 308\,000$ cm³ ▸ RECTANGULAR BOX (l = 60 cm, w = 50 cm, h = 70 cm) Volume = $l \times w \times h$ $= 60 \times 50 \times 70$ $= 210\,000$ cm³ Converting to litres (since 1 litre = 1 000 cm³): – Cylinder: 308 000 ÷ 1 000 = 308 litres

	– Rectangular box: 210 000 ÷ 1 000 = 210 litres The cylindrical drum holds 308 litres while the rectangular box holds only 210 litres — a difference of 98 litres, which is nearly 100 litres more maize! This confirms what the mill owner observed: the cylinder holds significantly more grain. Even though both containers look similar in size from the outside, the circular cross-section of the cylinder is mathematically more efficient at enclosing volume. This is why silos and storage drums used by farmers across the Rift Valley are almost always cylindrical — the shape maximises storage capacity for the material used.

Section 4: The Artisan's Design Challenge — Applying Mathematics to Optimise a Container

Imagine you are a metalsmith in Eldoret advising a posho mill owner who wants to build a new grain storage container. The owner has exactly 50 000 cm² of iron sheet. Using what you know about surface area and volume of solids (cylinder, rectangular prism, cone, sphere, or pyramid), explain how you would use mathematics to help the owner choose the best shape. You do not need to find the perfect answer — but you must use correct formulas, show reasoning, and justify your recommendation.	As a metalsmith's mathematics advisor, my job is to help the mill owner get the most storage capacity from 50 000 cm² of iron sheet. This means I want to maximise volume while keeping surface area fixed at 50 000 cm². I know from our investigations that different shapes enclose different volumes even with the same surface area. Here is how I would think through the options: ▸ OPTION 1 — Cylinder If I use a cylinder with r = 40 cm, I can find h by setting the surface area equal to 50 000 cm²: $2\pi r^2 + 2\pi rh = 50\ 000$ $2 \times (22/7) \times 1\ 600 + 2 \times (22/7) \times 40 \times h = 50\ 000$ $10\ 057.14 + 251.43h \approx 50\ 000$ $251.43h \approx 39\ 942.86$ $h \approx 158.9$ cm Volume ≈ $(22/7) \times 1\ 600 \times 158.9 \approx 799\ 771$ cm³ ≈ 800 litres ▸ OPTION 2 — Rectangular Prism (cube shape, l = w = h) For a cube: $6l^2 = 50\ 000 \rightarrow l^2 = 8\ 333.3 \rightarrow l \approx 91.3$ cm Volume = $91.3^3 \approx 760\ 586$ cm³ ≈ 761 litres ▸ RECOMMENDATION The cylinder gives more volume (~800 litres) than the cube (~761 litres) for the same 50 000 cm² of iron sheet. A sphere would give even more, but it is impractical for storing grain (it would roll and cannot stand upright). My recommendation to the mill owner: build a cylindrical drum. It is mathematically superior for maximising grain storage, it is stable and upright, and Kenyan metalsmiths have the tools and skills to bend iron sheet into cylindrical shapes efficiently. Mathematics does not just describe the world — it helps us design it better.

Section 5: Answering the Driving Question

Now answer the full driving question in your own words: Why does the same amount of metal sheet make containers that hold very different amounts of grain — and how can a Kenyan artisan use mathematics to design the best solid shape for storage or construction? Your answer should be a coherent paragraph or set of paragraphs that brings together everything you have learned across all 8 lessons. Reference surface area, volume, at least two solid shapes, and the Eldoret posho mill context.	The posho mill owner in Eldoret discovered a fundamental truth about three-dimensional geometry: surface area and volume are two different — and largely independent — properties of a solid. Surface area measures the total outer covering of a shape, telling us how much material like iron sheet is needed to build it. Volume measures the interior space, telling us how much grain, water, or other material the container can hold. Because these two quantities depend on shape in different ways, it is entirely possible — and in fact very common — for two containers built from the same area of metal sheet to hold very different amounts of grain. A cylinder and a rectangular box built from the same iron sheet will have different volumes because the formulas that govern them are different. The cylinder uses $\pi r^2 h$ for volume and $2\pi r^2 + 2\pi rh$ for surface area, while the rectangular prism uses lwh for volume and $2(lw + lh + wh)$ for surface area. The circular cross-section of a cylinder is a more 'efficient' shape — it encloses more area per unit of perimeter than a rectangle does. This principle, which mathematicians call the isoperimetric property, is why cylindrical containers consistently outperform rectangular ones in storage capacity when the same material is used. A Kenyan artisan can use this mathematics powerfully. Before cutting a single piece of iron sheet, the metalsmith can calculate the surface area needed for a given design, compare volumes of different shapes, and choose the geometry that best serves the client's needs. For a posho mill owner wanting maximum grain storage, a cylindrical drum is the mathematically optimal practical choice. For a construction project needing a decorative or structural roof, a pyramid or cone might be chosen for its aesthetic or load-bearing properties, with the artisan using surface area formulas to price the iron sheet precisely. Mathematics transforms guesswork into precision — and in the hands of a skilled artisan in Eldoret, Kisumu, or anywhere across Kenya, it becomes a tool for better design, less waste, and greater value.

RUBRIC			
Criterion	**Excellent (4)**	**Proficient (3)**	**Developing (1–2)**
Conceptual Understanding — Distinguishing Surface Area from Volume	Clearly and accurately defines both surface area and volume using correct mathematical language. Consistently explains why equal surface area does not imply equal volume, with explicit reference to the role of shape. Connects the concept directly to the Eldoret posho mill phenomenon.	Correctly defines surface area and volume and explains that shape affects the relationship between them. Makes reference to the posho mill context but explanation may lack full mathematical precision.	Shows partial understanding of either surface area or volume but confuses the two, or incorrectly states that equal surface area means equal volume. Limited or no connection to the phenomenon.
Mathematical Accuracy — Applying Correct Formulas and Calculations	Selects and correctly applies all relevant formulas (cylinder, rectangular prism, and at least one other solid). All calculations are accurate, fully shown step-by-step, and units are stated correctly throughout (cm^2, cm^3, litres where appropriate).	Applies correct formulas for cylinder and rectangular prism with mostly accurate calculations. Working is shown but may contain minor arithmetic errors. Units are mostly correct.	Attempts to use formulas but selects incorrect ones or makes significant calculation errors. Working is incomplete or units are missing or incorrect.
Evidence-Based Reasoning — Using Data to Support Claims	All claims about which container is better or which shape is optimal are directly supported by calculated numerical evidence. Comparisons are explicit (e.g., '308 litres vs 210 litres, a difference of 98	Most claims are supported by numerical evidence from calculations. Some comparisons are made but may not be fully explicit or may omit specific figures.	Makes claims about containers or shapes without sufficient numerical evidence, or evidence presented does not clearly support the conclusion drawn.

	litres'). Reasoning clearly flows from evidence to conclusion.		
Real-World Application — Connecting Mathematics to the Artisan's Context	Fluently connects mathematical findings to the real-world context of a Kenyan artisan or posho mill owner. Explains how the mathematics would practically guide design decisions (material planning, shape choice, cost). Uses Kenya-relevant context (e.g., Eldoret, grain storage, metalsmith) naturally and accurately throughout.	Connects calculations to the posho mill or artisan context in most sections. Practical implications are stated but may be general rather than specific to Kenyan contexts.	Little or no connection between the mathematics and the real-world context. The Eldoret phenomenon is mentioned but not meaningfully used to ground the explanation.
Communication — Clarity, Structure, and Mathematical Vocabulary	Writing is clear, logically organised, and uses correct mathematical vocabulary throughout (surface area, volume, cross-section, formula, optimise, capacity, dimensions, etc.). The driving question is answered fully and coherently in the final section, bringing together all prior sections into a unified explanation.	Writing is generally clear and organised. Correct mathematical vocabulary is used in most places. The driving question is answered but the final response may not fully integrate all prior sections.	Writing is unclear or poorly organised in places. Mathematical vocabulary is limited, missing, or used incorrectly. The driving question response is incomplete or does not draw on evidence from earlier sections.